CHAPTER 12 · REFERENCE DEPTH

Navigation Aids

The longest and most technical chapter in the book. Every system here is an application of the propagation and frequency principles of Chapters 7–10. We build each aid from first principles — how the signal is generated, how the receiver extracts navigation information, what its errors are, and how it is flown — then test it with worked examples and a full question bank.

SYLLABUS MAP

Part II (iv) Radio-navigation aids · Part I (viii) navigation facilities & principal frequencies

◎ Learning objectives — by the end of this chapter you will be able to…

- Explain the four kinds of navigation information radio provides (bearing, distance, vertical guidance, position) and the physical principle behind each.
- Describe the operating principle, frequency band, presentation, range and full error set of NDB/ADF, VOR, ILS and DME — and work numerical bearing/range problems.

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- Explain SSR Modes A/C/S, the interrogation/reply pulse scheme, garbling, fruit and side-lobe suppression, and the squawk codes.
- Contrast primary and secondary radar and state the basic pulse-radar relationships.
- Describe GNSS ranging, the need for four satellites, the error budget, DOP, and the augmentation systems (ABAS/SBAS/GBAS) including GAGAN and NavIC.
- Explain PBN (RNAV vs RNP), ACAS/TCAS logic, the radio altimeter, GPWS/EGPWS and INS/IRS.

12.0 How radio creates navigation information

12.1 NDB & ADF

12.2 VOR

12.3 ILS

12.4 DME

12.5 SSR, transponders & Mode S

12.6 Primary radar & radar principles

12.7 GNSS — GPS, GAGAN, NavIC

12.8 PBN — RNAV & RNP

12.9 ACAS / TCAS

12.10 Radio altimeter & GPWS/EGPWS

12.11 INS / IRS

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Modern navigation displays integrate data from VORs, DME, GNSS, and Inertial systems into a seamless picture.

12.0 How radio creates navigation information

THE UNIFYING IDEA

Every navigation aid in this chapter answers one of four questions, and each does so by exploiting a single measurable property of a radio wave:

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The Four Pillars of Navigation Aids

Question	Physical principle	Aids that use it
Which direction? (bearing)	Compare the **phase** or **amplitude** of signals to extract an angle, or find the null of a directional antenna	ADF (null), VOR (phase), ILS (modulation balance)
How far? (distance)	Measure the **time** a pulse takes to travel out and back; distance = ½ · c · t	DME, primary radar, radio altimeter
Where exactly? (position)	**Range from several known points** (multilateration / satellite ranging)	GNSS, DME/DME, INS (dead reckoning)

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Question	Physical principle	Aids that use it
Am I high/low, left/right of path?	Balance of two overlapping **modulated lobes**	ILS localizer & glide path, PAR

TWO FOUNDATIONAL NUMBERS

Radio waves travel at c = 3 × 10&sup8; m/s. A pulse therefore takes about **12.36 µs to travel one nautical mile and back** (the "radar mile"). This single fact underlies DME, radar and the radio altimeter. And bearing systems all reduce to measuring an **angle** — by phase, by amplitude balance, or by an antenna null.

CLASSIFYING ANY AID IN THE VIVA

For any aid, be ready to state four things in order: **(1) band, (2) principle, (3) what it gives the pilot, (4) its main errors/limitations.** Examiners build almost every nav-aid question from this skeleton.

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12.1 NDB & ADF

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The ADF dial in the cockpit points directly to the NDB ground station, providing a relative bearing.

12.1.1 The ground station (NDB)

IN PLAIN TERMS

The Non-Directional Beacon is the simplest radio aid: a ground transmitter that radiates equally in **all** directions (hence "non-directional"). It does nothing clever itself — all the navigation intelligence is in the aircraft's receiver, which determines the *direction* from which the beacon's signal arrives.

NDB — THE FACTS

Band: LF/MF, ICAO range **190–1750 kHz** (most are 200–415 kHz).

Propagation: primarily **ground (surface) wave**, which follows the Earth's curvature —

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giving useful range beyond line-of-sight at low level. **Emission:** a carrier keyed with a **two- or three-letter Morse ident** (NON A1A keyed carrier, or NON A2A where a 1020 Hz tone is added so the ident is heard on a standard AM receiver).

CLASSES & RANGE

Range depends on transmitter power and surface conductivity. As a rule of thumb: **Locator (L)** beacons (low power, used on approaches) ≈ 10–25 NM; **en-route** NDBs ≈ 50–100+ NM. Range is greater over **sea** (high conductivity) than over dry land or mountains.

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12.1.2 The airborne receiver (ADF) — how the bearing is found

FIRST PRINCIPLES — THE LOOP & SENSE ANTENNA

The Automatic Direction Finder uses a **loop antenna** whose reception pattern is a **figure-of-eight** (a "polar diagram" with two lobes and two sharp **nulls** at right angles to the lobes). A null is used rather than a peak because a null is far sharper and easier to detect precisely. But a single figure-of-eight has **two** nulls 180° apart — so the loop alone cannot tell whether the station is ahead or behind (180° ambiguity).

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To resolve this, the receiver adds an omnidirectional **sense antenna** (a circular pattern). Combining the loop's figure-of-eight with the sense antenna's circle produces a **cardioid** (heart-shaped) pattern with a **single** null. The receiver electronically rotates this pattern (a goniometer/servo) until it finds that single null, and drives the **ADF needle** to point at the station.

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WHAT THE NEEDLE SHOWS — RELATIVE VS MAGNETIC BEARING

On a fixed-card indicator the needle gives the **Relative Bearing (RB)** — the angle to the station measured *from the aircraft's nose*. To get the **Magnetic Bearing TO the station (QDM)** you add the heading:

Magnetic Bearing TO (QDM) = Relative Bearing + Magnetic Heading

(QDR = QDM ± 180°)

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On an **RMI** (Radio Magnetic Indicator) the compass card is slaved to the heading, so the needle reads the magnetic bearing (QDM) directly off the head of the needle.

WORKED EXAMPLE — ADF BEARING

Aircraft magnetic heading **040°**; the ADF relative bearing reads **030°** (i.e. 30° to the right of the nose).

QDM (magnetic bearing to the NDB) = RB + HDG = 030 + 040 = **070°**.

QDR (magnetic bearing from the NDB) = 070 − 180 = **250°**.

If the sum exceeds 360°, subtract 360. E.g. HDG 350°, RB 040° → QDM = 390 − 360 = **030°**.

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12.1.3 ADF/NDB errors — the full set

ADF Errors

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Error	Cause	When worst
Night effect	After dusk the ionosphere returns a **sky wave** that contaminates the ground wave; the two interfere and the needle wanders	Dusk & dawn, at longer ranges
Coastal (shoreline) refraction	The ground wave changes speed crossing the coast and **bends towards the coast**	Signal crossing coast at a shallow angle; near the coastline
Quadrantal error	The aircraft's own structure **re-radiates** the signal, biasing the loop	When the station is off the four "quadrantal" headings; reduced by calibration
Thunderstorm (static) effect	A CB's electrical discharges radiate strongly in LF/MF; the needle **points at the storm**	Near active CBs
Mountain / terrain effect	Reflections from high ground	Mountainous areas
Station interference	At night, distant co-channel beacons are heard via sky wave	Night

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⚠ EXAM TRAP

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NDB/ADF is the **least accurate** common aid (system accuracy roughly **±5°**). Always confirm the **Morse ident** before using a beacon, and treat a wandering needle near dusk, coast or a CB with suspicion. Unlike VOR, the NDB gives **no failure flag** — a needle that has quietly drifted can mislead you.

12.2 VOR — VHF Omnidirectional Range

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The Horizontal Situation Indicator (HSI) or traditional VOR dial allows the pilot to select and fly specific radials based on phase comparison.

12.2.1 The phase-comparison principle

FIRST PRINCIPLES — TWO 30 HZ SIGNALS

A VOR transmits **two** 30 Hz signals and lets the receiver compare their **phase difference** to determine the radial. One is a **reference** phase that is the *same in every direction*; the other is a **variable** phase whose timing depends on the bearing of the receiver from the station.

In a **conventional VOR (CVOR)**: the reference 30 Hz is transmitted as **frequency modulation** on a 9960 Hz sub-carrier (the same everywhere), while the variable 30 Hz is produced as **amplitude modulation** by physically/electronically **rotating a directional pattern at 30 revolutions per second**. The two signals are arranged to be **exactly in phase when the aircraft is due magnetic NORTH** of the station. As you move around the station, the variable phase slips relative to the reference by exactly the bearing angle — so the measured phase difference **is** the magnetic radial.

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VOR — THE FACTS

Band: VHF **108.0–117.975 MHz** (108.00–111.95 shared with ILS — VOR uses the frequencies with an **odd** first decimal e.g. 108.10; ILS uses even, e.g. 108.10 vs 108.15… check the chart). **Propagation:** line-of-sight (space wave). **Gives:** magnetic **radial TO or FROM** — 360 radials like spokes. **Ident:** 3-letter Morse, sometimes with voice.

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DVOR — WHY MOST MODERN STATIONS ARE DOPPLER

The **Doppler VOR (DVOR)** *reverses* the roles (reference = AM, variable = FM produced by Doppler shift from a large ring of antennas electronically "rotated"). The wide antenna ring makes DVOR far **less sensitive to site/terrain reflections**, so it is more

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accurate and easier to site. The cockpit indication is identical — the pilot cannot tell CVOR from DVOR.

12.2.2 Reading the instrument — OBS, CDI & the TO/FROM flag

IN PLAIN TERMS

You select a course with the **OBS (Omni Bearing Selector)**. The **CDI** (Course Deviation Indicator) needle shows how far left/right of that selected course you are, and the **TO/FROM** flag shows whether flying the selected course would take you *toward* or *away from* the station. Each dot of CDI deflection ≈ **2°**; full-scale (5 dots) ≈ **10°**.

WORKED EXAMPLE — CDI INTERPRETATION

You select OBS **090°** and get a **TO** flag with the needle **2 dots left**.
TO flag → flying 090° would take you to the station, so the station is roughly **east** of you (you are west of it, near the 270° radial).
Needle 2 dots left → the selected course is to your **left**; fly left (turn toward the needle) to intercept. 2 dots ≈ **4°** displacement from the 090° course line.
Golden rule: with a FROM flag the CDI senses correctly when your heading roughly matches the OBS; with a TO flag and the OBS set to the reciprocal, beware **reverse sensing**.

12.2.3 VOR errors & range

VOR Errors

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Error / limitation	Detail
Site & terrain (scalloping)	Reflections cause the needle to oscillate ("scalloping"); much reduced on DVOR
Cone of confusion	Directly overhead the station the signal is ambiguous — the TO/FROM flag flickers and the needle is unreliable for a short time
Line-of-sight range	VHF — range grows with altitude; blocked by terrain
Aggregate accuracy	≈ **±5°** system (ground station ±1–2°), far better than NDB; a **failure flag** appears if the signal is unusable

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VOR/DME, TACAN & VORTAC

A VOR is frequently co-located with a **DME** (VOR/DME) so a single station gives both bearing and distance — a complete fix. **TACAN** is the military equivalent (bearing + DME in UHF); a **VORTAC** co-locates a VOR with a TACAN so civil aircraft get VOR bearing + TACAN's DME.

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12.3 ILS — Instrument Landing System



The ILS uses intersecting localizer and glide path radio beams to provide precision guidance down to the runway threshold.

IN PLAIN TERMS

The ILS is a **precision approach** aid giving the pilot both **azimuth** (left/right of the centreline) and **vertical** (above/below the glide path) guidance down to low height, plus

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distance checkpoints. It is built from three independent radio systems.

12.3.1 The localizer — azimuth guidance

⚖ FIRST PRINCIPLES — THE 90/150 HZ BALANCE

An antenna array at the **far end** of the runway radiates two overlapping lobes along the approach: the left side carries a tone modulated at **90 Hz**, the right side at **150 Hz**. On the exact centreline the two tones are received with **equal depth of modulation** — the **Difference in Depth of Modulation (DDM) is zero** — and the needle centres. Off to one side, one tone dominates and the needle deflects. The receiver literally measures the **balance** between two modulation tones.

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LOCALIZER — FACTS

Band: VHF **108.10–111.95 MHz** (odd-tenth + 0.05 frequencies). **Course sector:** adjusted so full-scale deflection corresponds to roughly **±2.5°** (course width ≈ runway-width dependent, typically 3–6° total). **Sensitivity:** about **4× more sensitive** than a VOR for the same needle movement — a small deviation moves the needle a lot.

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⚠ BACK COURSE & REVERSE SENSING

The localizer also radiates a **back course** off the approach end. Flying inbound on the back course (or a front course outbound) gives **reverse sensing** — the needle moves opposite to the correction needed — unless the system/HSI compensates. Know this exists.

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12.3.2 The glide path — vertical guidance

GLIDE PATH — FACTS

Band: UHF **329–335 MHz** (automatically paired with the localizer frequency — you tune one box). **Principle:** same 90/150 Hz balance but in the **vertical** plane — **90 Hz dominates above** the path, **150 Hz below**. **Nominal angle: 3°**; usable half-width ≈ ±0.7°.

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FALSE GLIDE PATHS

The glide-path antenna pattern produces additional lobes, creating **false glide slopes** at higher angles (e.g. around 6°, 9°). They can be captured from above. Defence: intercept the glide path **from below** at the correct altitude after establishing on the localizer, and cross-check altitude/distance.

12.3.3 Marker beacons & ILS categories

Marker Beacons

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Marker	Tone / code	Light	Typical position
Outer (OM)	400 Hz, low tone, dashes	Blue	≈ 3.9–5 NM (final approach fix)
Middle (MM)	1300 Hz, alternate dot-dash	Amber	≈ 0.5–0.6 NM (near CAT I DH)

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Marker	Tone / code	Light	Typical position
Inner (IM)	3000 Hz, high tone, dots	White	Near threshold (CAT II/III)

ALL MARKER BEACONS SHARE 75 MHZ

Every marker — outer, middle, inner — transmits on a **75 MHz** carrier (VHF), radiating a narrow fan *upward*; the aircraft passes through it. Markers are increasingly replaced by **DME** for continuous distance. ICAO categories define the minima:

ILS Categories

Category	Decision Height	RVR (typical)
CAT I	≥ 200 ft	≥ 550 m
CAT II	100–200 ft	≥ 300 m
CAT IIIA	< 100 ft (or none)	≥ 200 m
CAT IIIB	< 50 ft (or none)	50–200 m
CAT IIIC	None	None (zero/zero)

AUTHOR CHECK BEFORE PUBLISHING

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Confirm exact CAT II/III DH/RVR values against the current DGCA CAR/ICAO Annex 10 & Annex 6, as national thresholds vary slightly. Also confirm protected-area (LSA/GSA critical & sensitive area) wording you wish to include.

12.4 DME — Distance Measuring Equipment

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⏱ FIRST PRINCIPLES — TIMING A PULSE PAIR

The aircraft (the **interrogator**) transmits **pulse pairs** on a UHF frequency. The ground **transponder** receives them, waits a **fixed 50 µs system delay**, and replies with pulse pairs on a frequency offset by **63 MHz**. The airborne set measures the **total round-trip time**, subtracts the 50 µs, and converts the remaining time to distance using the radar mile (≈12.36 µs per NM round trip). Because the aircraft initiates and the reply is uniquely jittered, each aircraft recognises only its *own* replies.

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DME — FACTS

Band: UHF **962–1213 MHz**. **Gives: slant range** (direct line-of-sight distance, not ground distance). **Capacity:** a ground transponder serves **≈ 100 aircraft**; beyond that it sheds the weakest (beacon saturation). Usually paired with a VOR or ILS so one selection drives both.

WORKED EXAMPLE — SLANT VS GROUND RANGE

You are at **6000 ft** (≈ 1 NM) overhead-ish a DME, reading **6 NM slant range**.
Ground distance = √(slant² − height²) = √(6² − 1²) = √35 ≈ **5.92 NM**.
Directly overhead (ground = 0) the DME reads your **height in NM**: 6000 ft ≈ 1 NM, so it reads ≈ 1 NM, not zero. **Slant-range error is greatest close-in and high**, and negligible far out at low level.

COCKPIT REALITY — THE DME ARC

Holding a constant DME distance while turning lets you fly a **DME arc** around a station — a common way to join a final approach. The DME's continuous distance is also what replaces marker beacons on a modern ILS ("4 DME", "2 DME" checks against altitude).

12.5 SSR, transponders & Mode S

FIRST PRINCIPLES — INTERROGATION & REPLY

Unlike primary radar (which listens for a passive echo), Secondary Surveillance Radar is a **cooperative** system. The ground interrogator transmits on **1030 MHz**; the aircraft **transponder** detects the interrogation and actively **replies on 1090 MHz** with coded data. Because the reply is a strong, coded transmission rather than a faint echo, SSR gives reliable **identity and altitude** with far less power.

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MODES & PULSE SCHEME

Mode A (identity): interrogation pulses P1–P3 spaced **8 µs**; the reply is a 4096-code (octal) squawk framed between framing pulses **F1/F2 (20.3 µs apart)**. **Mode C** (altitude): P1–P3 spaced **21 µs**; reply encodes pressure altitude in 100 ft steps (Gillham code). **Mode S** (Select): each aircraft has a unique **24-bit address**, so the radar interrogates aircraft individually, carries a data-link, and eliminates the garbling problem. ADS-B (1090ES) is built on the Mode S reply.

SSR Problems and Solutions

Problem	What it is	Cure
Garbling	Two aircraft at similar range/bearing reply overlapping in time	Mode S selective addressing
FRUIT	Replies triggered by *other* interrogators received by this radar	Defruiters; Mode S
Side-lobe replies	Transponder replies to the radar's antenna side-lobes, giving false bearings	**Side-Lobe Suppression (SLS)** — a P2 control pulse

SQUAWK CODES — MEMORISE EXACTLY

7500 unlawful interference (hijack) · **7600** radio/communications failure · **7700** general emergency. Also: **7000** conspicuity (VFR, many regions), **2000** entering from a non-

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SSR area, and the **SPI / IDENT** pulse when ATC asks you to "squawk ident".

MNEMONIC

"75 taken alive · 76 nix the comms · 77 going to heaven."

12.6 Primary radar & radar principles



Primary Surveillance Radar (PSR) transmits high-power pulses and listens for the faint, passive echo reflected off the aircraft skin.

IN PLAIN TERMS

Primary Surveillance Radar (PSR) is the original radar: transmit a pulse, listen for the **reflected echo** from the target, and time it. It needs **no equipment aboard** the aircraft, but gives only **range and bearing** — no identity, no altitude.

THE PULSE-RADAR RELATIONSHIPS

Range = ½ · c · t (echo time there-and-back). **PRF** (pulse repetition frequency) sets the maximum unambiguous range — too high a PRF and a distant echo returns after the next pulse is sent (range ambiguity). **Pulse width** sets minimum range and range resolution; **beam width** sets bearing resolution. **MTI/Doppler** processing rejects stationary clutter to show moving targets.

Primary Radar Types

Radar	Role
ASR / TAR	Airport/terminal surveillance (approach control)
ARSR	En-route (area) surveillance — long range
SMR	Surface Movement Radar — ground traffic in poor visibility
PAR	Precision Approach Radar — controller-guided azimuth + elevation talkdown (Ch 17)

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Radar	Role
Weather radar	Airborne, ~9.3 GHz (X-band), detects precipitation/CB

12.7 GNSS — GPS, GAGAN & NavIC

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The GAGAN system uses geostationary satellites to broadcast corrections and integrity messages, augmenting the core GNSS signals over the Indian Flight Information Region.

12.7.1 How satellite ranging fixes a position

FIRST PRINCIPLES — PSEUDO-RANGING & THE FOURTH SATELLITE

Each satellite broadcasts its position and a precisely-timed code. The receiver measures the **time of flight** of the code → a **range** to that satellite, placing the aircraft on a sphere around it. Two satellites narrow it to a circle, three to two points — so in principle **3 satellites give a 3-D position**. But the receiver's own clock is cheap and slightly wrong, and a small clock error scales by c into a large range error. A **fourth satellite** is used to solve for that clock error as well — hence **four satellites are required for a 3-D fix.**

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GPS — ARCHITECTURE

Space segment: ≥ 24 satellites in **6 orbital planes** at ≈ **20,200 km**, period ≈ **11 h 58 min**, inclination 55°. **Signals: L1 = 1575.42 MHz** (also L2 1227.6, L5 1176.45 MHz). **Control segment:** ground monitor/upload stations. **User segment:** the receiver. Other constellations: **GLONASS** (Russia), **Galileo** (EU), **BeiDou** (China), **NavIC/IRNSS** (India, regional — L5 & S-band, ~7 satellites).

12.7.2 Error budget, DOP & integrity

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GNSS Errors

Error source	Note
Ionospheric / tropospheric delay	Largest natural error; partly modelled, dual-frequency cancels ionosphere

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Error source	Note
Satellite clock & ephemeris	Corrected by the navigation message
Multipath	Reflections off ground/structures near the antenna
Receiver noise	Internal
Geometry (GDOP/PDOP)	Satellites **spread wide** across the sky give a strong fix (low DOP); satellites bunched together give a weak fix (high DOP)

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AUGMENTATION — MAKING GNSS GOOD ENOUGH FOR APPROACHES

ABAS (Aircraft-Based, e.g. **RAIM** — Receiver Autonomous Integrity Monitoring): the receiver cross-checks redundant satellites to detect a faulty one. **SBAS** (Satellite-Based): geostationary satellites broadcast wide-area corrections + integrity — **GAGAN** (India), WAAS (USA), EGNOS (Europe), enabling APV/LPV approaches. **GBAS** (Ground-Based, GLS): a local airport station for precision approaches.

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GAGAN — THE INDIAN SBAS

GAGAN (GPS Aided GEO Augmented Navigation) uses geostationary satellites to broadcast corrections and an **integrity** message over the Indian region, improving GPS accuracy to a few metres and enabling approach procedures with vertical guidance.

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Integrity — the system's ability to warn the pilot promptly when it should not be used — is what makes augmentation essential for IFR approaches.

12.8 PBN — RNAV & RNP

IN PLAIN TERMS

Performance-Based Navigation (PBN) frees aircraft from flying directly station-to-station: instead they fly defined paths to a stated **accuracy**. The key distinction the exam wants:

RNAV VS RNP — THE ONE-LINE DIFFERENCE

RNAV = Area Navigation: fly any path within the coverage of the nav aids, to a stated lateral accuracy. **RNP** = RNAV **plus on-board performance monitoring and alerting** — the system continuously checks it is meeting the required accuracy and **warns the crew if it is not**. That self-monitoring is the whole difference.

PBN Specifications

Specification	Lateral accuracy (NM, 95%)	Phase
RNAV 10 / RNP 10	10	Oceanic / remote

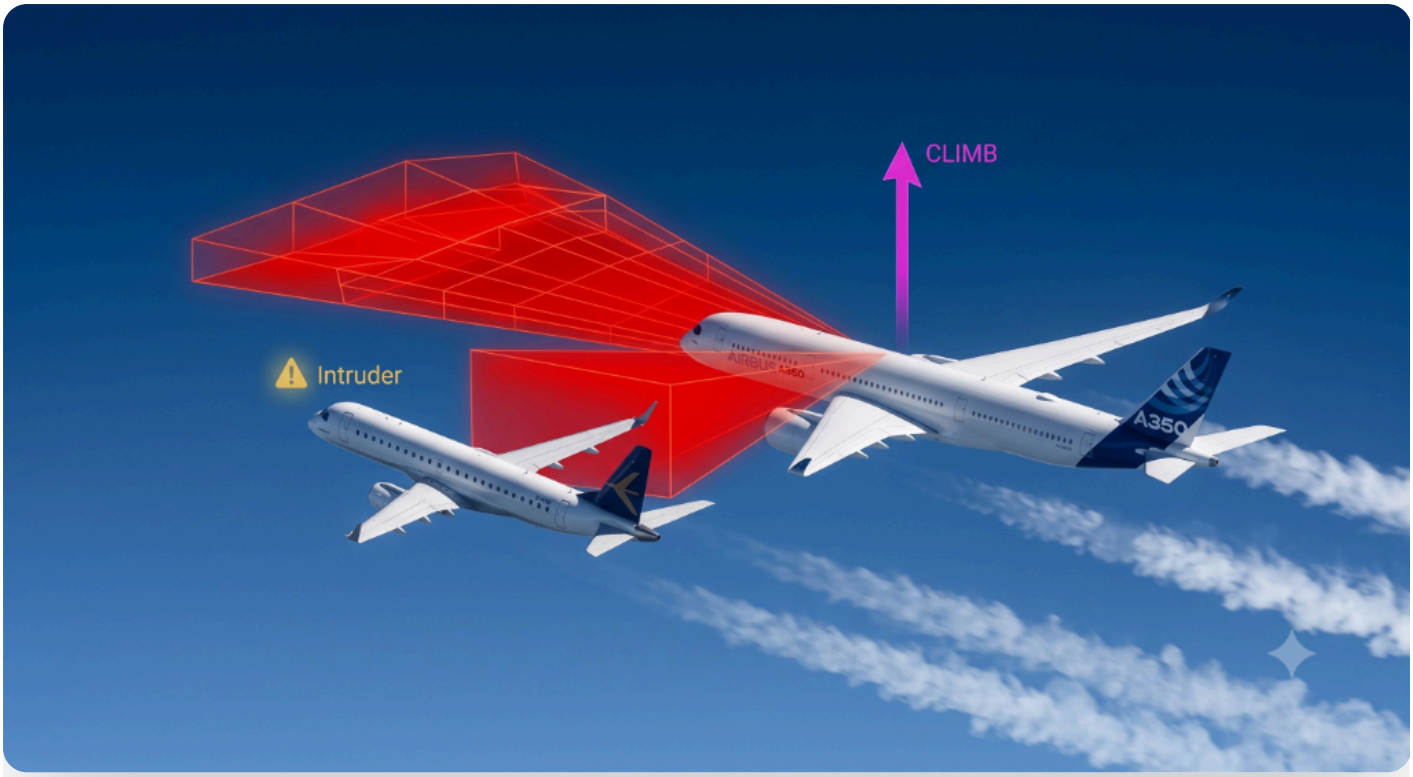
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Specification	Lateral accuracy (NM, 95%)	Phase
RNAV 5	5	En-route continental
RNAV 1 / 2	1 / 2	Terminal
RNP 4	4	Oceanic (with monitoring)
RNP 1	1	Terminal (with monitoring)
RNP APCH / AR	0.3 → lower	Approach (AR = authorisation required, curved paths)

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12.9 ACAS / TCAS

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When a conflict is detected, TCAS II issues a coordinated Resolution Advisory (RA) commanding a vertical manoeuvre, such as "CLIMB", to ensure safe separation.

FIRST PRINCIPLES — USING TRANSPONDERS TO SEE TRAFFIC

ACAS (the ICAO term; TCAS II is the common implementation) **interrogates the transponders** of nearby aircraft — exactly like a tiny airborne SSR — and tracks their range, closure and altitude. It builds a picture independent of ground radar and warns of conflicts **in the vertical plane**.

TA THEN RA

Traffic Advisory (TA): "TRAFFIC, TRAFFIC" — **≈ 40 s** to closest approach; alerts the crew to look out, but commands nothing. **Resolution Advisory (RA): ≈ 25 s** to closest approach — a **vertical command** (climb/descend/maintain/level-off). When both aircraft have TCAS II, they **coordinate via Mode S** so one climbs and the other

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descends. **An RA is followed even if it conflicts with an ATC instruction**, and the crew advises ATC ("TCAS RA").

⚠ LIMITATIONS

TCAS only sees aircraft with an operating **transponder** (it cannot see non-transponding traffic). RAs are **vertical only** — never turn in response to an RA. It is a last-resort safety net, not a substitute for see-and-avoid or ATC separation.

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12.10 Radio altimeter & GPWS/EGPWS



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The radio altimeter measures true height above the surface, feeding the GPWS which issues immediate audio and visual warnings (e.g., "TERRAIN, PULL UP") when closure rates become dangerous.

↓ RADIO (RADAR) ALTIMETER

Band: ≈ 4200–4400 MHz (SHF). **Technique: FM-CW** (frequency-modulated continuous wave) — the frequency difference between the transmitted and reflected signal is proportional to height. **Measures true height above the surface directly below (AGL)**, typically **0–2500 ft**, used on approach and for auto-land. It reads terrain height, not a barometric level.

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⚠ GPWS MODES (1–7) & EGPWS

The **Ground Proximity Warning System** uses radio-altimeter height and aircraft data to warn of dangerous situations: **1** excessive descent rate · **2** excessive terrain closure rate · **3** altitude loss after take-off · **4** unsafe terrain clearance (not in landing config) · **5** below glide slope · **6** altitude callouts / excessive bank · **7** wind shear. **EGPWS (Enhanced)** adds a worldwide **terrain database** and GPS position for **forward-looking** terrain alerting — warning of terrain ahead, not just below.

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12.11 INS / IRS

🧭 FIRST PRINCIPLES — NAVIGATION WITH NO OUTSIDE SIGNAL

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An Inertial Navigation System measures the aircraft's own **accelerations** (accelerometers) and **rotations** (gyroscopes), and mathematically integrates them from a **known start position** to compute present position continuously — needing **no external radio signal at all.** It is therefore completely self-contained and un-jammable.

PLATFORM VS STRAPDOWN; DRIFT; HYBRIDISATION

Older systems use a gimballed stable **platform**; modern **strapdown** IRS fix the sensors to the airframe and use **ring-laser or fibre-optic gyros**. Because errors accumulate through integration, the position **drifts** with time (typically **≈ 2 NM/hr**). **Schuler tuning** (an 84.4-minute period) prevents the system oscillating with vehicle motion. In practice the IRS is **blended with GNSS** (hybrid IRS/GPS) so GNSS bounds the drift while inertial provides smooth, high-rate, jam-resistant data.

☆ Numbers to memorise

Essential Facts for Chapter 12

Aid	Band / frequency	Principle	Gives / key figures
NDB / ADF	LF/MF 190–1750 kHz	Loop null + sense → cardioid	Relative bearing; ±5°; ground wave

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Aid	Band / frequency	Principle	Gives / key figures
VOR	VHF 108.0–117.975 MHz	30 Hz phase comparison (ref FM / var AM)	Magnetic radial; ±5°; in-phase at North
ILS LOC	VHF 108.10–111.95 MHz	90/150 Hz DDM balance (azimuth)	Centreline; ±2.5° full scale
ILS GP	UHF 329–335 MHz	90 above /150 below (vertical)	3° path; false slopes ~6°/9°
Markers	75 MHz (OM 400/MM 1300/IM 3000 Hz)	Vertical fan	Distance checkpoints
DME	UHF 962–1213 MHz	Pulse-pair timing; 50 µs delay; 63 MHz offset	Slant range; ~100 a/c; radar mile 12.36 µs/NM
SSR	1030 ↑ / 1090 ↓ MHz	Interrogate–reply; Mode A 8 µs / C 21 µs / S 24-bit	ID + altitude; 7500/7600/7700
GPS	L1 1575.42 MHz	Satellite pseudo-ranging	4 sats for 3-D; ~20,200 km; GAGAN augments

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Aid	Band / frequency	Principle	Gives / key figures
Radio altimeter	4200–4400 MHz	FM-CW	True AGL 0–2500 ft
INS/IRS	(no radio)	Accelerometers + gyros	Self-contained; drift ~2 NM/hr; Schuler 84.4 min

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🖉 Question bank

Part A — MCQs (click an option to check)

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1. The ADF resolves the 180° ambiguity of the loop antenna by:

Using a higher frequency

Adding a sense antenna to form a cardioid with a single null

Rotating the aircraft

Comparing 90 and 150 Hz tones

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Answer: Adding a sense antenna to form a cardioid with a single null.

The omnidirectional sense antenna combines with the figure-of-eight loop to give a cardioid with one null.

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2. Heading 350°(M), ADF relative bearing 040°. The QDM is:

310°

030°

050°

210°

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Answer: 030°.

QDM = RB + HDG = 040 + 350 = 390 − 360 = 030°.

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3. A VOR's reference and variable 30 Hz signals are in phase when the aircraft is:

Overhead the station

Due magnetic north of the station

On the 180° radial

At the cone of confusion

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Answer: Due magnetic north of the station.

By design the phase difference is zero on the 360° (north) radial; it then equals the radial.

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4. In a conventional VOR, the reference phase signal is carried as:

30 Hz AM

30 Hz FM on a 9960 Hz sub-carrier

A 90 Hz tone

A Morse code

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Answer: 30 Hz FM on a 9960 Hz sub-carrier.

CVOR: reference = FM on 9960 Hz sub-carrier (same in all directions); variable = AM from the rotating pattern.

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5. The ILS localizer indicates centreline when:

90 Hz dominates

150 Hz dominates

The depth of modulation of 90 and 150 Hz is equal (DDM = 0)

The DME reads zero

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Answer: The depth of modulation of 90 and 150 Hz is equal (DDM = 0).

On the centreline the two tones are received equally — zero difference in depth of modulation.

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6. ILS marker beacons all transmit on:

329–335 MHz

108.10–111.95 MHz

75 MHz

1030 MHz

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Answer: 75 MHz.

Outer/middle/inner markers share a 75 MHz carrier, differing in tone and code.

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7. A false glide slope is typically encountered:

Below the true 3° path

Above the true path (e.g. ~6°), if intercepted from above

On the localizer back course

Only at night

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Answer: Above the true path (e.g. ~6°), if intercepted from above.

Antenna side-lobes create false slopes at higher angles; intercept the GP from below to avoid them.

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8. DME measures distance by:

Comparing two 30 Hz phases

Timing the round trip of pulse pairs, less a 50 µs ground delay

Measuring received signal strength

Counting satellites

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Answer: Timing the round trip of pulse pairs, less a 50 µs ground delay.

The interrogator times the reply (minus the fixed 50 µs transponder delay) to derive slant range.

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9. At 12,000 ft (2 NM) directly overhead a DME, the indicator reads approximately:

0 NM

2 NM

6 NM

12 NM

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Answer: 2 NM.

Overhead, ground range is zero so slant range equals height: 12,000 ft ≈ 2 NM.

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10. An SSR transponder is interrogated on, and replies on, respectively:

1090 / 1030 MHz

1030 / 1090 MHz

329 / 335 MHz

75 / 75 MHz

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Answer: 1030 / 1090 MHz.

Ground interrogates on 1030 MHz; aircraft replies on 1090 MHz.

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11. Side-Lobe Suppression in SSR prevents:

Garbling

FRUIT

Replies to the antenna's side lobes giving false bearings

Altitude errors

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Answer: Replies to the antenna's side lobes giving false bearings.

A P2 control pulse stops the transponder replying when interrogated via a side lobe.

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12. The transponder code for radio communications failure is:

7500

7600

7700

7000

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Answer: 7600.

7500 hijack, 7600 radio failure, 7700 emergency.

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13. Which radar requires equipment carried aboard the aircraft to function?

Primary radar

Secondary radar

Both

Neither

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Answer: Secondary radar.

SSR needs a transponder; primary radar works on the passive echo from any target.

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14. A 3-D GNSS position fix requires a minimum of how many satellites?

2

3

4

6

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Answer: 4.

Three for position plus one to solve the receiver clock error = four.

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15. The GPS L1 carrier frequency is:

1227.60 MHz

1575.42 MHz

1090 MHz

1176.45 MHz

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Answer: 1575.42 MHz.

L1 = 1575.42 MHz (L2 = 1227.6, L5 = 1176.45).

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16. GAGAN is best described as:

A ground-based primary radar

A satellite-based augmentation system (SBAS) for the Indian region

India's own satellite constellation for global coverage

An inertial system

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Answer: A satellite-based augmentation system (SBAS) for the Indian region.

GAGAN broadcasts corrections + integrity over India via GEO satellites.

(NavIC/IRNSS is the regional constellation.)

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17. The essential difference between RNAV and RNP is that RNP adds:

A higher cruising altitude

On-board performance monitoring and alerting

A second VOR

Mode S

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Answer: On-board performance monitoring and alerting.

RNP = RNAV plus the requirement that the system monitors its own accuracy and alerts the crew.

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18. A TCAS Resolution Advisory commands a manoeuvre in:

The horizontal plane (turn)

The vertical plane (climb/descend)

Any plane

Speed only

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Answer: The vertical plane (climb/descend).

RAs are vertical-only; never turn in response to an RA.

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19. The radio altimeter operates around 4200–4400 MHz and indicates:

Pressure altitude

True height above the surface directly below (AGL)

Altitude above mean sea level

Slant range to a beacon

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Answer: True height above the surface directly below (AGL).

FM-CW radio altimeter gives AGL, typically 0–2500 ft.

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20. A characteristic of INS used alone is that its position error:

Is constant regardless of time

Grows with time (drift), typically a few NM per hour

Depends on satellite geometry

Only occurs over the sea

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Answer: Grows with time (drift), typically a few NM per hour.

Integration accumulates error, so unaided INS drifts (~2 NM/hr); GNSS is used to bound it.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **Explain how an NDB and ADF together give a bearing, and list the main errors.**

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Model Answer:

The NDB radiates omnidirectionally in LF/MF on the ground wave. The airborne ADF loop has a figure-of-eight pattern with sharp nulls but a 180° ambiguity; a sense antenna combines with it to form a cardioid with a single null, which the receiver rotates onto the station to drive the needle to the relative bearing (QDM = RB + HDG). Main errors: night effect, coastal refraction, quadrantal error, thunderstorm effect, mountain effect and station interference; overall ≈ ±5° and no failure flag.

▼ ? **Describe the phase principle of a VOR.**

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Model Answer:

A VOR sends a reference 30 Hz (FM on a 9960 Hz sub-carrier, identical in all directions) and a variable 30 Hz (AM from a pattern rotating at 30 rev/s). The two are in phase due magnetic north; elsewhere the phase difference equals the magnetic radial. DVOR reverses the roles using Doppler for better siting. Accuracy ≈ ±5°, line-of-sight, with a cone of confusion overhead and a failure flag.

▼ ? How does the ILS provide azimuth and vertical guidance?

Model Answer:

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The localizer (VHF 108.10–111.95 MHz) radiates 90 Hz on one side and 150 Hz on the other; equal depth of modulation (DDM = 0) marks the centreline. The glide path (UHF 329–335 MHz) does the same vertically — 90 Hz above, 150 Hz below a 3° path. Marker beacons (75 MHz) give distance checkpoints. Beware false glide slopes above the true path and back-course reverse sensing.

▼ ? Why does DME read slant range, and when is the error significant?

Model Answer:

DME times the round trip of pulse pairs (less the 50 µs transponder delay), which measures the direct line-of-sight distance — the slant range. The error between slant and ground range is greatest when close to the station and high; overhead it reads the aircraft's height in NM. Far out at low level it is negligible.

▼ ? Explain SSR Modes A, C and S and the problems Mode S solves.

Model Answer:

Mode A returns an identity squawk (P1–P3 spaced 8 µs); Mode C returns pressure altitude (21 µs spacing); Mode S addresses each aircraft individually by a 24-bit code and carries a

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data link. Mode S solves garbling (overlapping replies) and reduces FRUIT, and underpins ADS-B. Side-lobe suppression (P2) prevents false replies via the antenna side lobes.

▼ ? Why are four satellites needed for a GNSS position, and what limits accuracy?

Model Answer:

Three satellites fix a 3-D position, but the receiver's clock error scales by the speed of light into a large range error, so a fourth satellite is needed to solve for that clock error. Accuracy is limited by ionospheric/tropospheric delay, satellite clock/ephemeris errors, multipath, receiver noise and satellite geometry (DOP). Augmentation (ABAS/RAIM, SBAS/GAGAN, GBAS) adds the accuracy and integrity needed for approaches.

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▼ ? What is the difference between RNAV and RNP?

Model Answer:

Both let an aircraft fly defined paths to a stated lateral accuracy, but RNP additionally requires on-board performance monitoring and alerting — the system continuously checks it is meeting the required navigation performance and warns the crew if it is not.

▼ ? How does TCAS work and what are its limitations?

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Model Answer:

TCAS interrogates nearby transponders to track range, closure and altitude, issuing a Traffic Advisory (~40 s) and, if needed, a Resolution Advisory (~25 s) commanding a vertical manoeuvre, coordinated with the other TCAS via Mode S. It only detects transponding aircraft, gives vertical-only RAs, and an RA is flown even against ATC, with ATC then advised.

▼ ? Distinguish primary and secondary radar.

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Model Answer:

Primary radar transmits a pulse and times the passive reflected echo, needing nothing aboard but giving only range and bearing. Secondary radar interrogates a cooperative transponder (1030 MHz) which replies (1090 MHz) with identity and altitude, using far less power and giving richer data.

▼ ? What does the radio altimeter measure and how, and what uses it?

Model Answer:

An FM-CW system around 4200–4400 MHz measuring true height above the surface directly below (AGL), typically 0–2500 ft. It feeds the GPWS/EGPWS (terrain warnings), auto-land and decision-height callouts on precision approaches.

Part C — Numerical problems (tap for worked solutions)

▼ P1. Heading 120°(M), ADF relative bearing 250°. Find QDM and QDR.

Solution:

QDM = 250 + 120 = 370 − 360 = **010°**; QDR = 010 + 180 = **190°**.

▼ P2. At 9000 ft you read 3 NM on the DME. What is the approximate ground distance?

Solution:

height ≈ 9000/6076 ≈ 1.48 NM; ground = $\sqrt{(3^2 - 1.48^2)} = \sqrt{(9 - 2.19)} = \sqrt{6.81} \approx$ **2.61 NM**.

▼ P3. A radar echo returns 1235 µs after transmission. What is the target range?

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Solution:

using 12.36 µs/NM round trip, range = 1235 / 12.36 ≈ **100 NM**.

▼ 🧮 **P4. You are on the 045° radial of a VOR, tracking inbound. What OBS and flag give correct sensing?**

Solution:

inbound track = 045 − 180 = **225°**; set OBS **225**, expect a **TO** flag, needle centred when on track.

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⚡ 60-SECOND REVISION CARD

ADF/NDB LF/MF, loop-null+sense→cardioid, RB (QDM=RB+HDG), ±5°, many errors, no flag.
VOR VHF, 30 Hz phase (ref FM/var AM), in-phase at North, radials, ±5°, cone of confusion, flag.
ILS LOC VHF 90/150 DDM; GP UHF 329–335 (90 above/150 below, 3°, false slopes); markers 75 MHz; CAT I–III.
DME UHF 962–1213, pulse timing, 50 µs delay, slant range, ~100 a/c.
SSR 1030↑/1090↓, Mode A/C/S, SLS/garble/FRUIT, 7500/7600/7700.
GNSS L1 1575.42, 4 sats for 3-D, DOP, augment ABAS/SBAS(GAGAN)/GBAS; NavIC=India constellation.
RNP=RNAV+monitoring/alerting. **TCAS** vertical RA, Mode-S coordinated. **RadAlt** 4200–4400 FM-CW AGL. **INS** self-contained, drifts ~2 NM/hr.

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